

ECE501: Digital Image Processing

Project Topic: Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

GROUP 19

Week 3 Progress Report

Date: October 5, 2026

Deliverable: Literature-backed pipeline design, dataset shortlist and evaluation plan

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Objectives of the Week

- Study published work on texture-based fabric defect detection using GLCM, LBP and Gabor features [1]–[7].
- Shortlist public fabric defect datasets with defect-free and defective images.
- Design the complete detection and localization pipeline: pre-processing, feature extraction, normal-texture reference, anomaly scoring, thresholding and post-processing.
- Define how detection and localization quality will be measured.

Work Done This Week

A. Literature study

We reviewed open-access papers on texture-based fabric inspection. Tajeripour et al. [1] show that LBP features computed on windows of a defect-free reference image, together with a threshold, can flag defective windows in both plain and patterned fabrics. GLCM-based methods [2], [3] treat the woven fabric as a texture image and use co-occurrence statistics such as energy, contrast, homogeneity and correlation to detect local defects. The lightweight deep-learning detector of Kang [4] reaches high accuracy and speed, but it needs labelled training data and more computing power. This supports our choice of simple, adaptive classical descriptors that learn only from defect-free fabric.

B. Dataset shortlisting

We shortlisted three public sources. AFID [8] offers images of several fabric types with pixel-level ground-truth masks, which suit localization metrics. The TILDA raw-fabric database and the Hong Kong Polytechnic University patterned-fabric database, both used in [4], provide further fabric and defect variety. The final choice will depend on availability of ground-truth masks and defect-free reference images.

C. Pipeline design

- Pre-processing: grayscale conversion, Gaussian or median denoising and CLAHE contrast adjustment, so that uneven lighting is not mistaken for a defect.
- Patch extraction: each image is divided into fixed-size patches (starting at 128×128 pixels, to be tuned), since window-based analysis is the basis of [1].
- GLCM features [5]: co-occurrence matrices at distances 1 and 2 and orientations 0° , 45° , 90° and 135° ; contrast, dissimilarity, homogeneity, energy and correlation, averaged over orientations.
- LBP features [6]: rotation-invariant uniform LBP histograms at ($P = 8$, $R = 1$) and ($P = 16$, $R = 2$).
- Gabor features [7]: a filter bank of four orientations and three frequencies; the mean and standard deviation of each response are used as features.
- Normal-texture reference: the mean and covariance of features from defect-free patches, one model per fabric type; each test patch is scored by its Mahalanobis distance from this reference.
- Adaptive threshold: $\tau = \mu + k\sigma$, computed from scores of defect-free patches (k to be tuned between 2 and 3), followed by morphological opening and closing and removal of very small regions.

D. Evaluation plan

Patch-level detection and pixel-level localization will be scored as shown in Table 1.

Level	Metric	Definition
Patch (detection)	Accuracy, Precision, Recall, F1	Does the patch contain a defect or not?
Pixel (localization)	IoU	$TP / (TP + FP + FN)$
Pixel (localization)	Dice	$2TP / (2TP + FP + FN)$

Table 1. Evaluation metrics.

Literature Review

Topic / Paper	Key Takeaway	Relevance to Our Project
Tajeripour et al., Fabric Defect Detection Using Modified Local Binary Patterns [1]	Builds an LBP reference from defect-free fabric, compares window histograms against it and uses a threshold to find defective windows; works on plain and patterned fabrics and is simple enough for online use.	Closest to our approach: a normal-texture reference, window analysis and a threshold.

Zhang and Fan, Fabric Defect Detection based on GLCM [2]	Treats woven fabric as a texture image and uses GLCM features to detect local defects.	Supports using GLCM as a core descriptor.
Real Time Fabric Defect Detection System on an Embedded DSP Platform [3]	Derives co-occurrence statistics such as energy, contrast, homogeneity and correlation for real-time fabric inspection.	Shows that GLCM statistics are cheap enough for fast inspection.
Kang, Research on fabric defect detection method based on lightweight network [4]	Improves YOLOv5s with attention and feature-enhancement modules and a CIoU loss; about 102 FPS, mAP@0.5 of 86.4% on raw and 75.8% on patterned fabric.	Shows what deep detectors achieve and what they cost; motivates our lightweight, adaptive classical pipeline.
Haralick et al., Textural features for image classification [5]	Defines the gray-level co-occurrence statistics used as texture descriptors.	Original source for our GLCM features.
Ojala et al., Multiresolution gray-scale and rotation invariant texture classification with LBP [6]	Introduces uniform, rotation-invariant LBP at several (P, R) scales.	Source of our two LBP scales.
Jain and Farrokhnia, Unsupervised texture segmentation using Gabor filters [7]	Uses multi-orientation, multi-frequency Gabor filter banks to separate textures.	Basis of our Gabor filter bank.
Silvestre-Blanes et al., A public fabric database for defect detection methods and results [8]	Public database of fabric images with defect classes and pixel-level ground-truth masks.	Candidate dataset for training references and evaluating localization.

Results and Analysis

This week was devoted to study and design, so the pipeline has not yet been run on the full data. Quantitative results (patch accuracy, precision, recall, F1, pixel IoU and Dice) will therefore be reported in Week 4 using the metrics defined in Table 1. Based on the literature, we expect GLCM and LBP to give a reasonable first detection result [1]–[3], and Gabor features to help with oriented defects such as broken ends and picks [7]. These are expectations to be tested, not findings.

Challenges Faced

- Dataset choice: public databases differ in fabric type, image size, defect classes and availability of pixel-level masks, and we need both defect-free images and ground truth.
- Patch size: small windows localize defects better but give unstable texture statistics; large windows do the opposite. We plan to test several sizes.
- Few normal samples: the feature vector can have more dimensions than there are defect-free patches, which makes the covariance matrix hard to invert. We plan to regularize it.
- Uneven illumination can look like a defect, which is why contrast normalization is included in pre-processing.
- Choosing the threshold factor k so that false alarms and missed defects are balanced.

Plan for Next Week (Week 4)

- Download the chosen dataset and implement the pre-processing and patch-extraction steps.
- Implement GLCM and LBP feature extraction, and Gabor features if time allows.
- Build the normal-texture reference, compute anomaly scores and apply the adaptive threshold.
- Evaluate with the metrics in Table 1 and save sample heatmaps and masks as figures.

Conclusion

In Week 3 we studied open-access work on GLCM, LBP and Gabor based fabric defect detection, shortlisted public datasets and designed the full adaptive pipeline together with its evaluation plan. The literature shows that a defect-free reference with a statistical threshold is a simple and effective idea [1], while deep models are accurate but heavier [4]. Week 4 will implement the pipeline and report the first quantitative results.

References

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